

String Manipulation

TIP101

Agenda

1 Welcome 🙌

2 String Manipulation

3 **BREAK**

4 Collaborative Problem Solving

5 Wrap Up

10 minutes

35 minutes

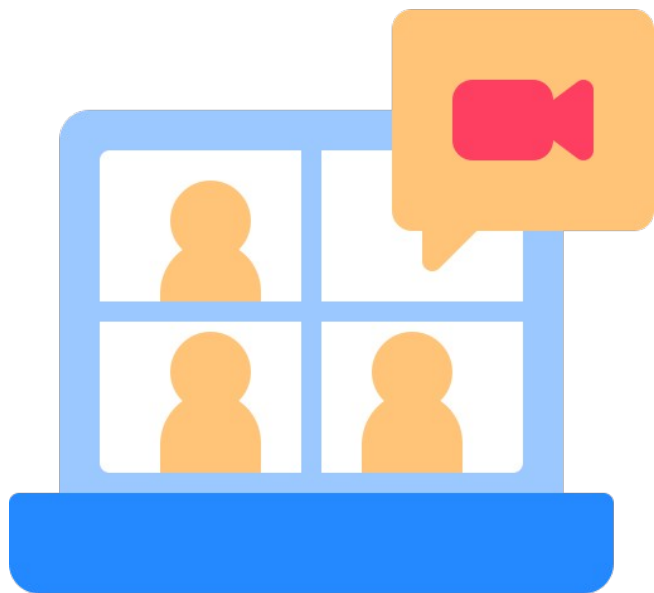
5 minutes

60 minutes

10 minutes

10 minutes

Welcome! 🖐️



**Turn on your
cameras**



**Fill out the entry
ticket**



Announcements & Reminders

* Assessment Reminder:

- Fix it garage sessions ended for now (onboarding specific to errors relating to VS Code, the course portal, Slack, etc.)
- Study hall sessions will remain active !
- Utilize VS Code debugging functionalities.



Icebreaker

5 minutes



Question:

You have been selected to give the next TED Talk on any topic of your choice. What are you choosing as your topic?



Answer:

Share your answer in the Zoom chat below. Multiple options are encouraged.

Your Feedback



Glows

- + Students specifically calling each other out for their support.
- + Appreciated seeing more built in functions.



Grows

- Still some mixed feedback on pacing (too fast vs too slow).
- Jumping between content (dictionary->strings)

35 minutes

String Manipulation

String Basics (Reminder)

- * Strings are sequences of characters.
- * Common operations: concatenation, slicing, and formatting.
- * Useful methods: `.lower()`, `.upper()`, `.split()`, `.replace()`.

```
s = "Hello World"
print(s.lower())  # "hello world"
print(s[:5])     # "Hello"
```

String Formatting (two ways)

- * String formatting allows you to create a string “template” that can dynamically fill in placeholders in the string. For example, something like this string which will contain user specific data:
 - “{userName} has signed in at {currentTime}.”

```
userName = "Cory2026"  
currentTime = "March 10th, 8:15pm"  
print(f"{userName} has signed in at {currentTime}.")
```

String Formatting (two ways)

- * String formatting allows you to create a string “template” that can dynamically fill in placeholders in the string. For example, something like this string which will contain user specific data:
 - “{userName} has signed in at {currentTime}.”

```
userName = "Cory2026"  
currentTime = "March 10th, 8:15pm"  
print("{} has signed in at {}".format(userName, currentTime))
```



Instructor Demo

Given a string `s` containing words separated by spaces, reverse the order of the words. Remove extra spaces and ensure that words are separated by a single space in the output.

```
def reverse_words(s):  
    pass
```

Input: `s = " the sky is blue "`

Output: `"blue is sky the"`



Instructor Demo

Given a string `s` containing words separated by spaces, reverse the order of the words. Remove extra spaces and ensure that words are separated by a single space in the output.

```
def reverse_words(s):  
    pass
```

Input: `s = " the sky is blue "`

Output: `"blue is sky the"`

Challenge!

Sometimes in interviews, you will be given a specific restriction or requirement.

What if the interviewer says that you may not use the `.reverse()` method?

Break Time!

Take 5 mins to step away from the computer. Feel free to turn off your camera and return promptly after 5 mins.



60 minutes

Collaborative Problem Solving



Breakout Rooms

60 minutes

1. 🏃 Navigate to your breakout room.
2. 🙌 Introduce yourselves!
3. 🧑💻 Work on the problems for your selected unit.

10 minutes

Wrap Up



One Good Thing

Let's reflect on today's session. Maybe you got a question answered, overcame a specific coding challenge, or learned how to do something completely new!

In the chat, share at least one good thing you've learned or experienced during the session today.

Let's celebrate together!



EARLY-COURSE SURVEY



1. **Open** the email you enrolled in CodePath with
2. **Locate** the message from support@codepath.org titled:

[Course Code] Early Course Survey - Please Complete
3. **Open** your unique survey link
4. **Answer** each question completely and honestly
5. **Celebrate:** You're helping CodePath courses improve!



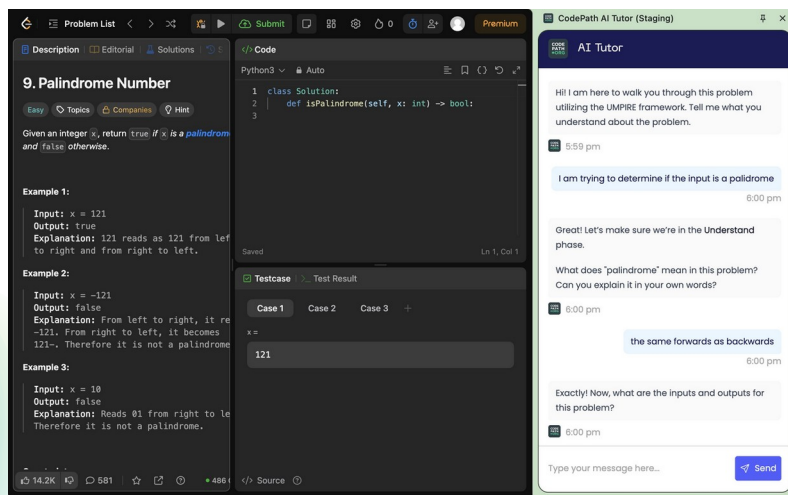
Final Reminders

- ☐ Finish the problem sets for additional practice
- ☐ Take the weekly HackerRank assessment by Sunday @ midnight
- ☐ Visit a **Study Hall Session** to deepen your understanding, work through challenges, and connect with course material alongside your peers!
- ☐ Remember to check out the **Codepath AI Tutor Bot** and earn an absence back! [→](#)

How do I get the most out of this course?



CodePath AI Tutor



Build your skills with support! Take the following steps:

- * Install the AI Tutor on chrome: <http://go.codepath.org/tiptutor>
- * Solve 2 LeetCode problems on 2 different days

Once you've taken the steps above, unlock a new opportunity!

- * +1 absence back for completion(one-time offer)

[Take a look at the FAQ if you have any questions](#)